



Australian Government
Bureau of Meteorology

Tide Calendar 2 Manual

Tidal Unit, Bureau National Operations Centre



Revision history

Date	Version	Description	Author
10/09/2020	0.1	First draft completed	James Chittleborough

Release history

Date	Version	Status	Audience	Approval
			Internal	

Document management register

Document	File reference
	NTC-2020-01

Review Status

Reviewer	Date Reviewed	Version Reviewed
Mike Davis, NTC Data Controller	xx/xx/2020	0.1

Release Signatories

Approval	Name	Signature	Date

Contents

1	Introduction	4
1.1	Source-code	4
1.2	Executable.....	4
1.3	Execution.....	4
1.4	Output Files	4
2	Frequently Asked Questions	5
3	Cookbook.....	6
3.1	Example 1. Default contemporary format.....	7
3.2	Example 2. ..add markers on the tide curve to denote monthly and yearly high and low tides.....	9
3.3	Example 3. ..add markers adjacent tide times to denote monthly and yearly high and low tides	11
3.4	Example 4. ..use the greyscale colorscheme	13
3.5	Example 5. ..use standard color colorscheme.....	15
3.6	Example 6. ..use custom colorscheme.....	17
3.7	Example 7. ..used filledcurve rather than impulses	19
3.8	Example 8. ..add shading for the night-time period between sunset and sunrise.....	21
3.9	Example 9. ..add a line tracing the tide curve.....	23
3.10	Example 10. ..highlight tides above/below certain thresholds	25
3.11	Example 11. ..add highlighting of tide times above/below certain thresholds ...	27
3.12	Example 12. ..float the key/legend	29
3.13	Example 13. Produce the Pacific Sea Level Monitoring project calendars	31
3.14	Example 14. Basic classic calendar	33

List of Tables

No table of figures entries found.

List of Figures

Figure 1. Names of the tide_calendar2 color variables..... 17

1 Introduction

This manual explains how to produce tide calendars using the program *tide_calendar2*, which succeeds the earlier generation program *tide_calendar*. For more basic tide tables, there is the program *tide_table2*.

1.1 Source-code

The application sourcecode is located as follows

Production (shared storage): `/g/ns/ntc/op/dm/bin/tidecalendars2/`

Development (ntcm server): `/home/james/bin/utilities/tidecalendars2/`

1.2 Executable

The *tide_calendar2* executable is symbolically linked from the data management *bin* directory on the shared storage device `/g/ns/ntc/op/dm/bin`. Do not run from this directory (see section 1.3)

1.3 Execution

From any sensible directory of your choice, type *tide_calendar2* on the command line with appropriate options (assuming you have the `/g/ns/ntc/op/dm/bin` path in your profile)

Use *tide_calendar2* or *tide_calendar2 -h* for help, which will list the available options (see section 3).

Use *tide_calendar2 -H* to display this manual.

The *tide_calendar2* program will create a temporary local sub-directory with a random name of the form

*tmp.**

This temporary directory will be removed at the end of execution, unless you specify the *-k* keep option (or exit prematurely with Ctrl+C). The *-k* option is useful for debugging any problems.

1.4 Output Files

The *tide_calendar2* pdf output files are placed in the appropriate folder within the tidal realm repository, unless the *-o* option is used to specify an alternative output directory.

The naming convention of calendars is as follows

`<antt>_<year>_<tide type>.cal_<calendar type>_<timezone>.pdf`

For example: `61600_2021_hlw.cal_cal1_loc.pdf` indicates the 2021 high and low waters for Port Adelaide (ANTT 61600), presented as a calendar of the type `cal1` in local time.

2 Frequently Asked Questions

How can I apply my own custom colour scheme for a tide calendar?

Invoke a custom colour scheme using `-c 3` option. The custom colours can be changed by way of *tide_calendar2 -e*, which will allow you to edit the CUSTOM_CALENDAR.dem file (noting at this stage it is a shared file). The naming conventions of the colour variables are shown in Figure 1.

How do I change the colour scheme for preset calendar types?

The preset calendar types (eg *pslm*) have had their colour scheme hard-wired and are not intended to be changed.

How do I change the range of the y-axis

The range of the y-axis is determined by *tide_calendar2* automatically by interrogating the prediction file. If this is inadequate then there is a need to speak to the developers.

How do I change the name of the station?

The name of the station is obtained from the *tide_register*.

How do I add a port to the system?

tide_calendar2 interfaces to ports registered in the tidal registry. Use the *tide_register* program to add a port to the registry (and *tide_realm* for performing tidal analysis and predictions)

The local government has introduced or discontinued Daylight Savings, how do I reflect this on the calendars?

Daylight savings is handled automatically by the operating system, provided the tzdata package is up to date and the correct `timezone.utc_offset` and `timezone.name` are registered in the tidal registry (eg UTC+09:30 and Australia/Adelaide)

I want to make some changes to the software because it is not working or not giving me what I need.

Development of *tide_calendar2* is undertaken in a separate development environment. It is preferred that changes are made there, and then migrated across. If it is absolutely necessary to make changes in the production environment (excluding operator-level configuration file CUSTOM_CALENDAR.dem) please ensure you document any changes and inform the developers. This way the software can be mirrored across development and production, else operator-level changes will be lost.

3 Cookbook

Usage: tide_calendar2 [Options] <year> <antt_list>

year is the required year of predictions

<antt_list> is a list of antt numbers

Options:

-t <type>	Specify tide calendar type (see Calendar types below)
-p <preset>	Specify preset type and options (see Presets below)
-z <zone>	Specify times in loc (local), dst (daylight saving) or utc (UTC) timezones (default is loc)
-m "<list>"	Create calendars only for those months given in list
-f	Use filledcurves (default is impulses)
-l	Trace the tide curve with a line
-x	Mark lowest/highest tides for the month/year (use -X to also place mark adjacent tide times)
-b	Put a white border around the markers
-w <wet>	Highlight tides above given wetting height
-d <dry>	Highlight tides below given drying height (use -W or -D to also highlight tide times)
-c <scheme>	Specify colorscheme as 0 (black/white), 1 (grey-scale), 2 (standard) or 3 (custom) (Default is 0)
-e	Edit custom gnuplot colorscheme
-s	Shade period between sunset and sunrise
-F	Float the key (default is place key in footer)
-i	Add an imperial scale in addition to metric
-O	Order pages for double-sided top-bound calendar (and place a please turn over message on june)
-k	Keep temporary data files (default is delete)
-o <dir>	Specify the output directory (default puts calendar in the tidal realm)
-M	Create individual monthly pdf files
-h	Command line help
-H	Display manual

Calendar types:

cal1	contemporary (default)
cal2	classic

Presets:

pslm	calendars for PSLGM project
------	-----------------------------

Examples:

3.1 Example 1. Default contemporary format

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2* <year> <antt>

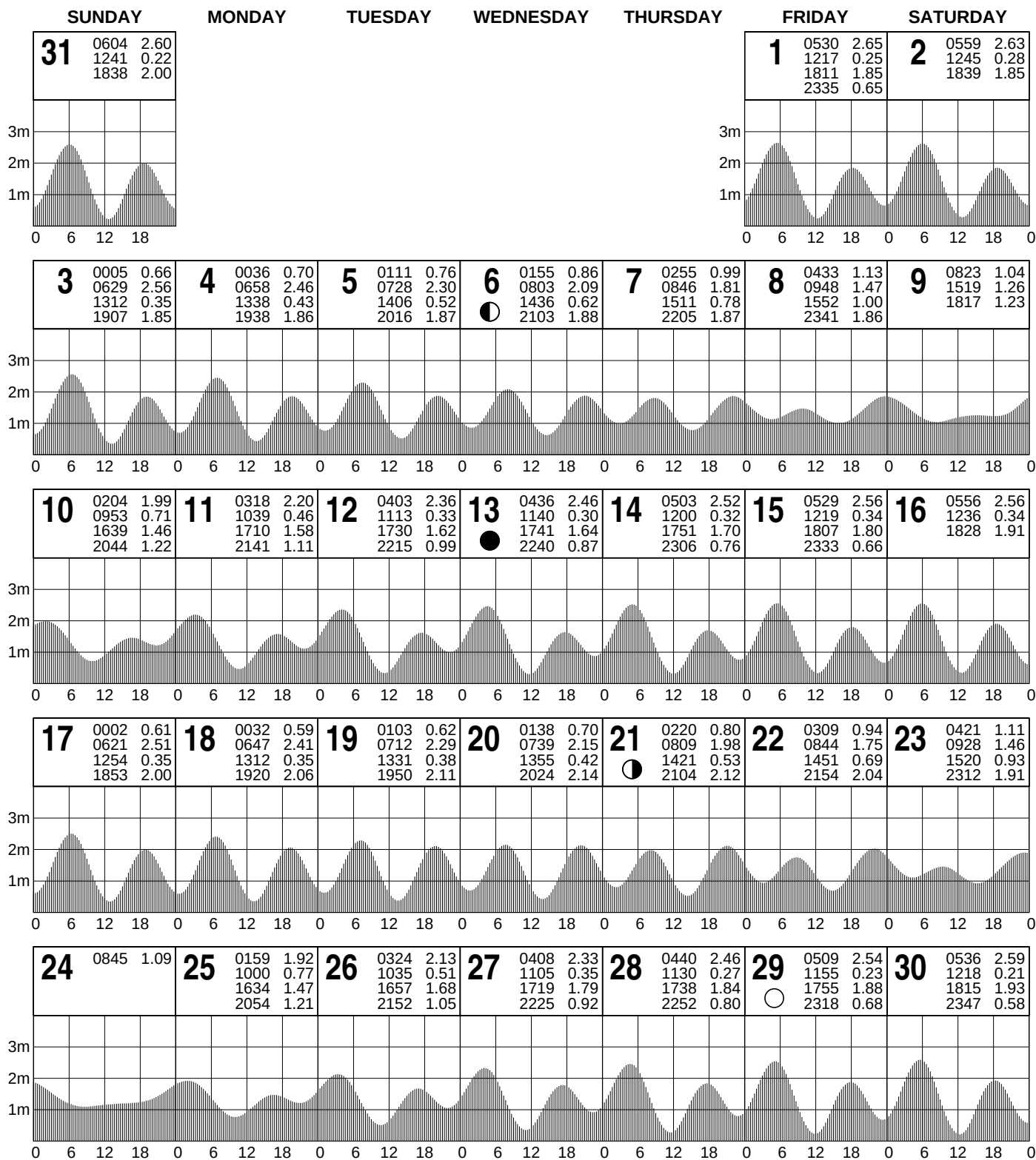
Notes: The default tide calendar is the contemporary type (cal1) with greyscale color scheme.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



- New moon
- ◐ First quarter
- Full moon
- ◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.2 Example 2. ..add markers on the tide curve to denote monthly and yearly high and low tides

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2 -x <year> <antt>*

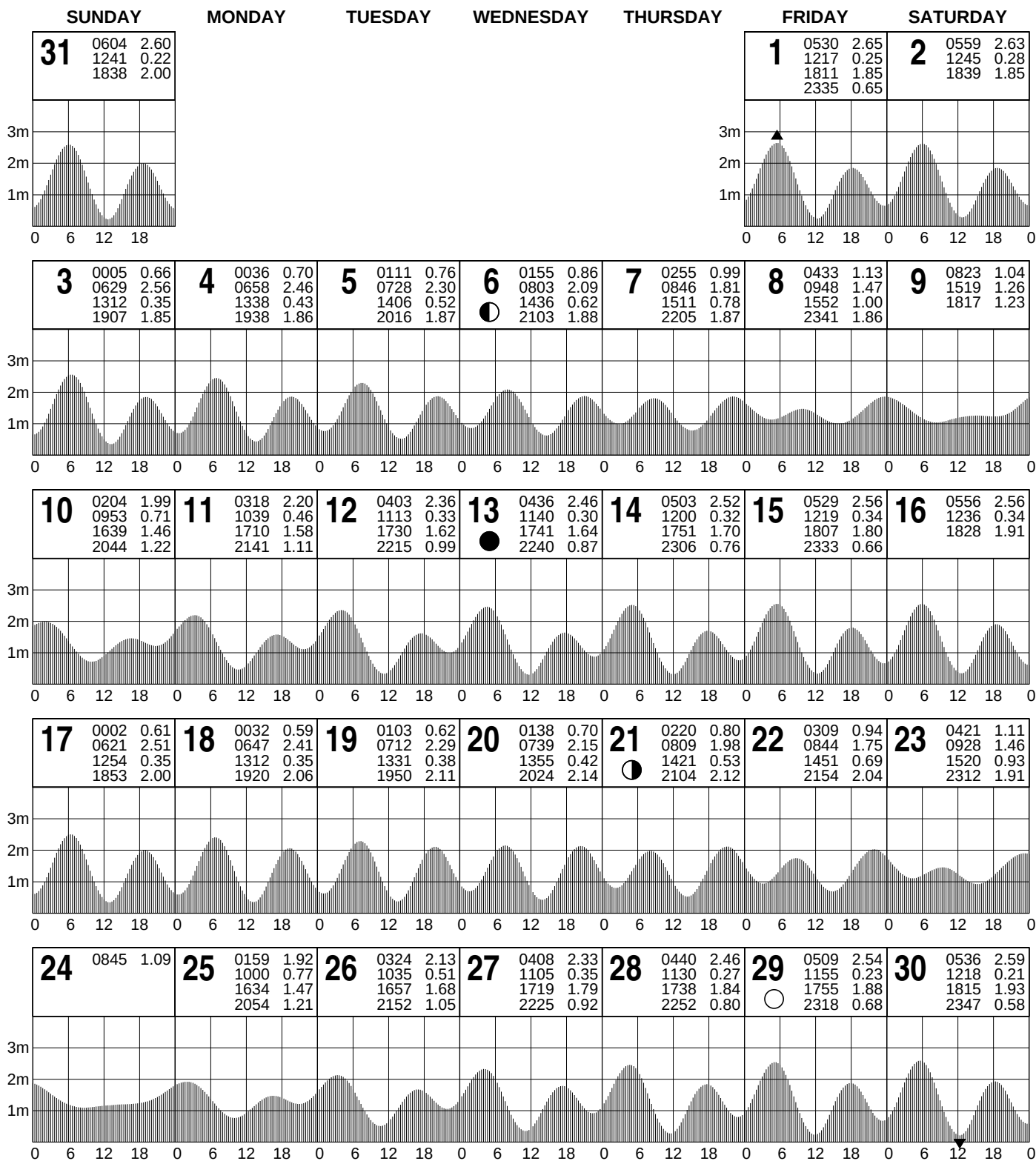
Notes: This option adds triangles adjacent the tide curve pointing upwards for highest tides and downwards for lowest tides.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month
▼ Lowest tide of the month

● New moon
◐ First quarter
○ Full moon
◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.3 Example 3. ..add markers adjacent tide times to denote monthly and yearly high and low tides

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2* -**X** <year> <antt>

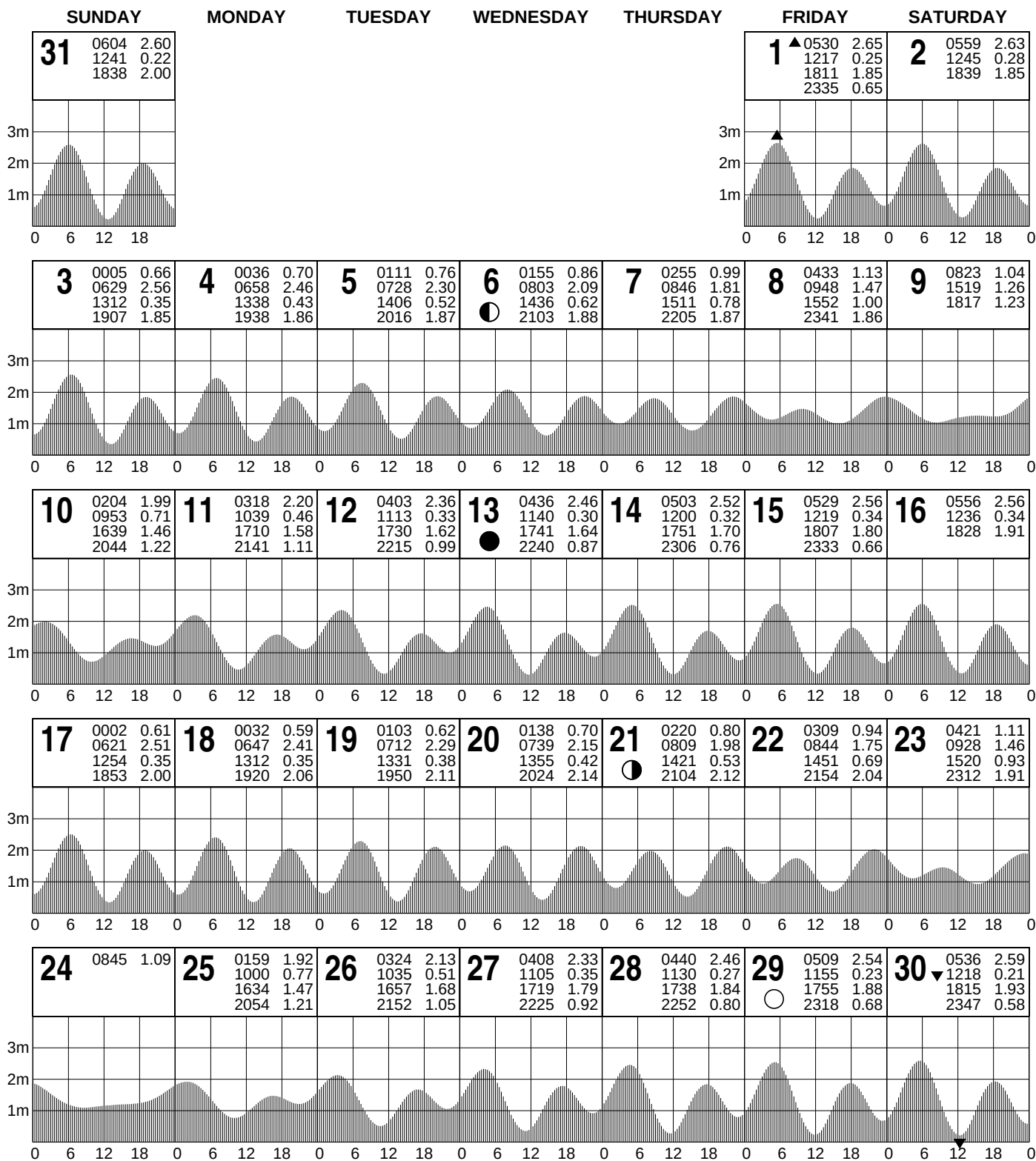
Notes: For more explicit marking of the time and height of monthly/yearly extreme tides.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month
▼ Lowest tide of the month

● New moon
◐ First quarter
○ Full moon
◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.4 Example 4. ..use the greyscale colorscheme

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2 -X -c 1 <year> <antt>*

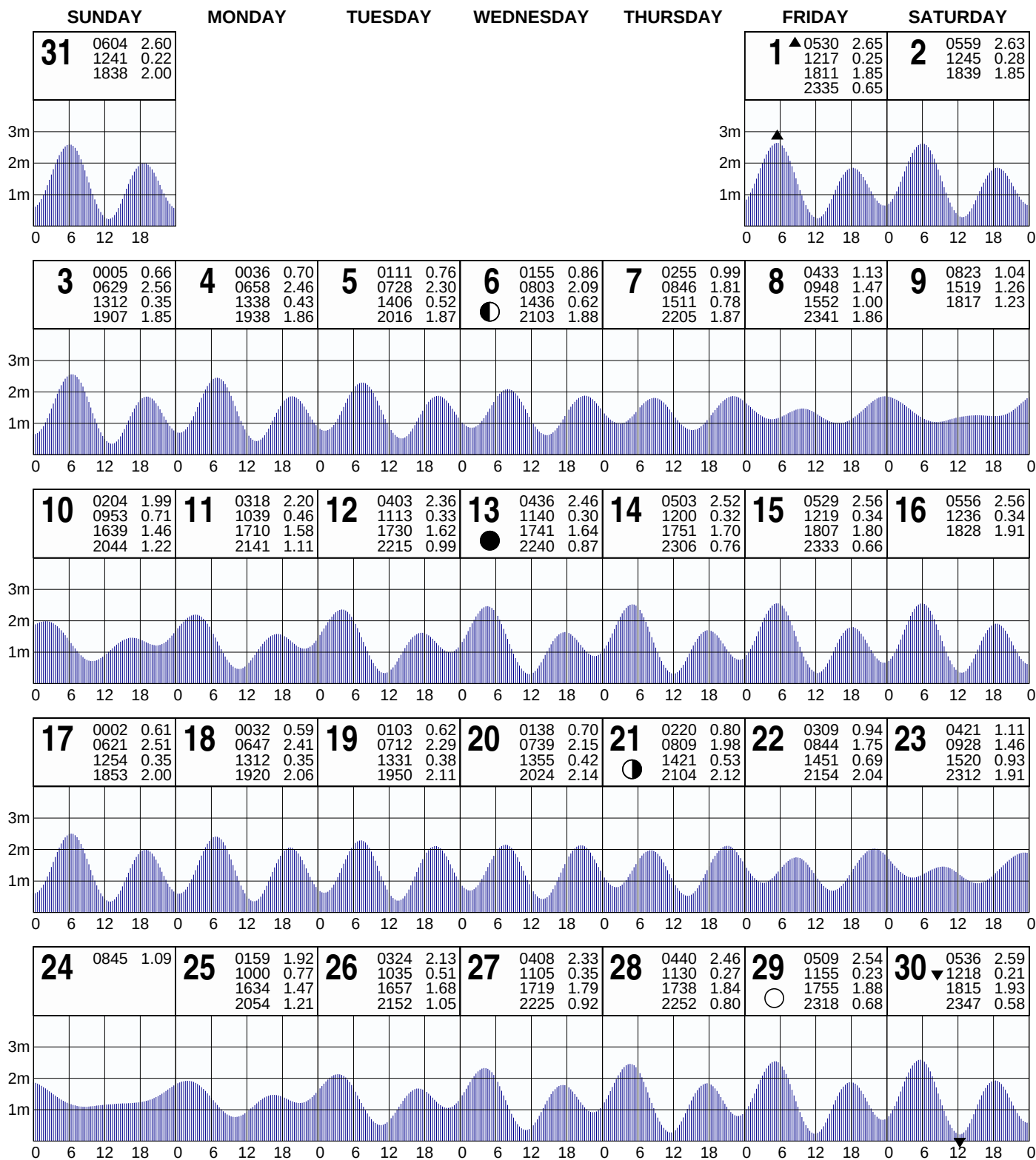
Notes: Choose greyscale by way of colorscheme option 1 (ie -c 1)

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month
▼ Lowest tide of the month

● New moon
◐ First quarter
○ Full moon
◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.5 Example 5. ..use standard color colorscheme

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2 -X -c 2 <year> <antt>*

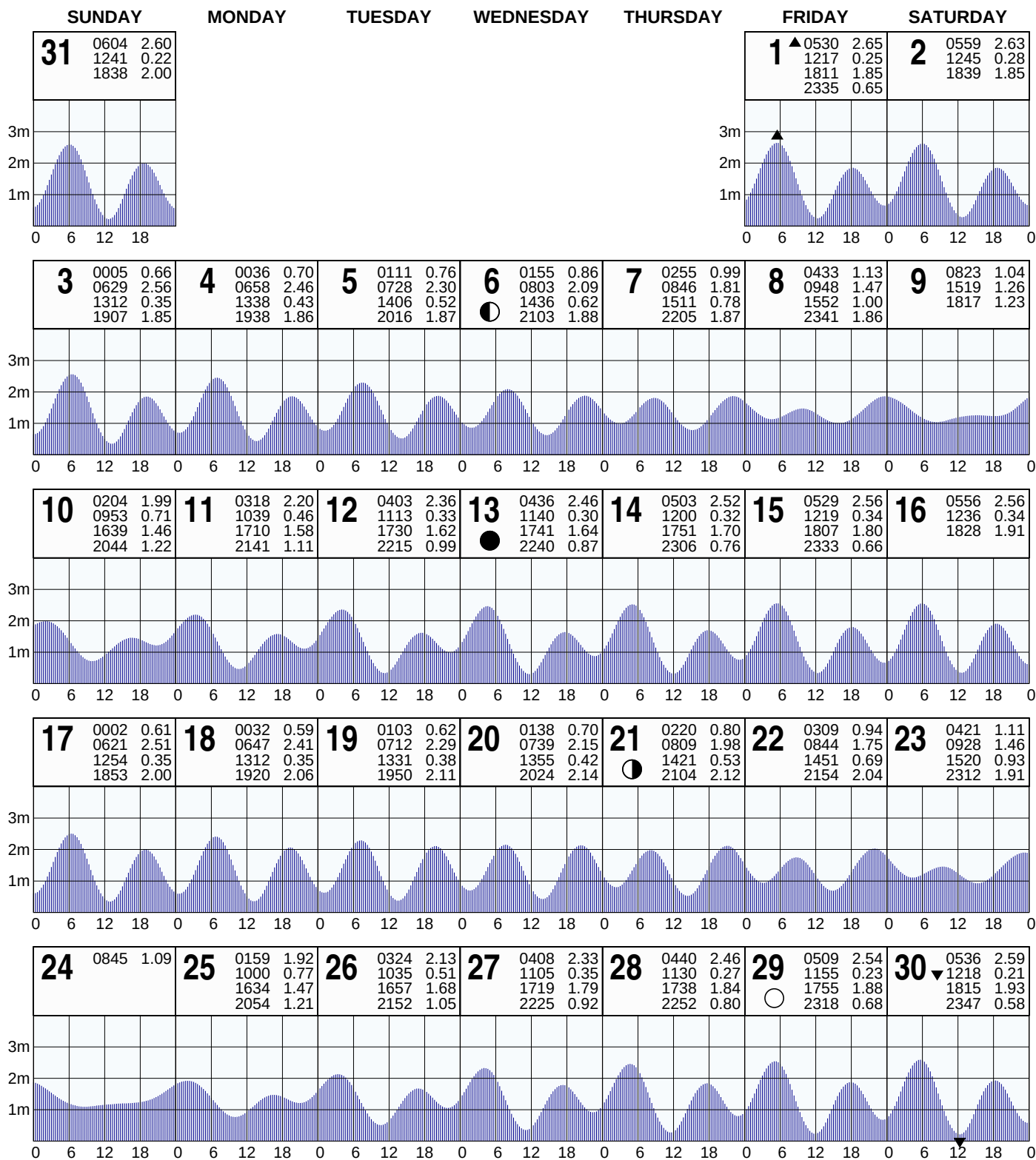
Notes: Choose standard color colorscheme by way of colorscheme option 2 (ie -c 2)

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month
▼ Lowest tide of the month

● New moon
◐ First quarter
○ Full moon
◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.6 Example 6. ..use custom colorscheme

CLASSIFICATION: UNRESTRICTED

Command: `tide_calendar2 -X -c 3 <year> <antt>`

Notes: Choose custom colorscheme by way of colorscheme option 3 (ie `-c 3`). The colors can be customised by way of editing the custom configuration, issuing `tide_calendar2 -e`

The names of the `tide_calendar2` color variables (whose values can be changed) are given in Figure 1.

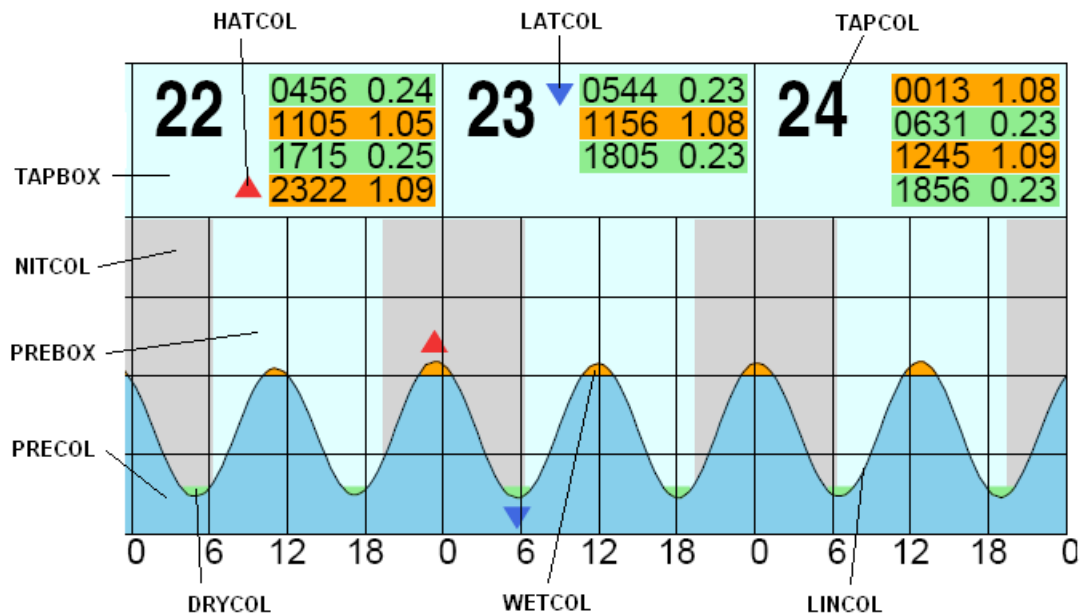


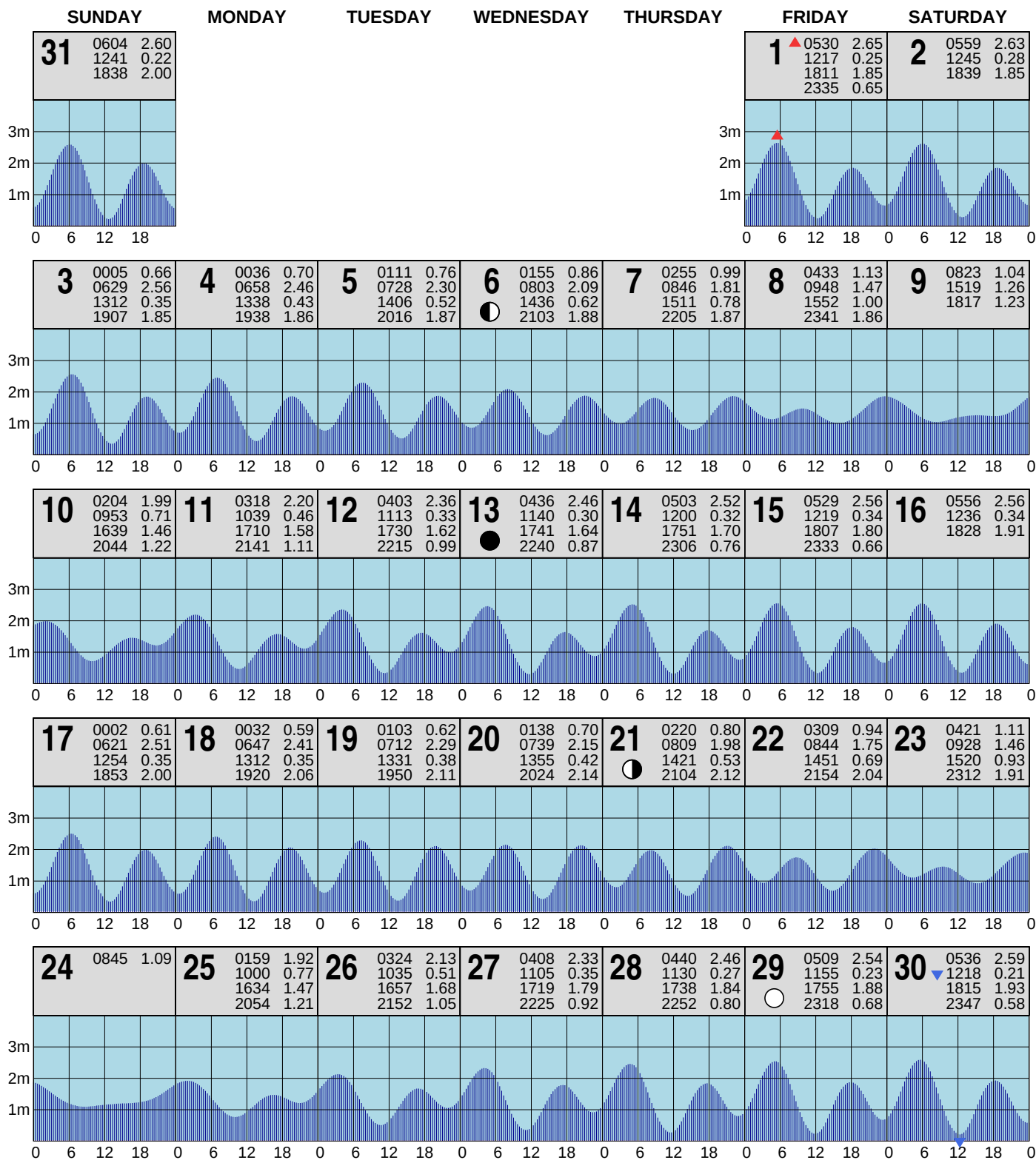
Figure 1. Names of the `tide_calendar2` color variables

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month
 ▼ Lowest tide of the month

● New moon
 ◐ First quarter
 ○ Full moon
 ◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.7 Example 7. ..used filledcurve rather than impulses

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2 -X- c 2 -f <year> <antt>*

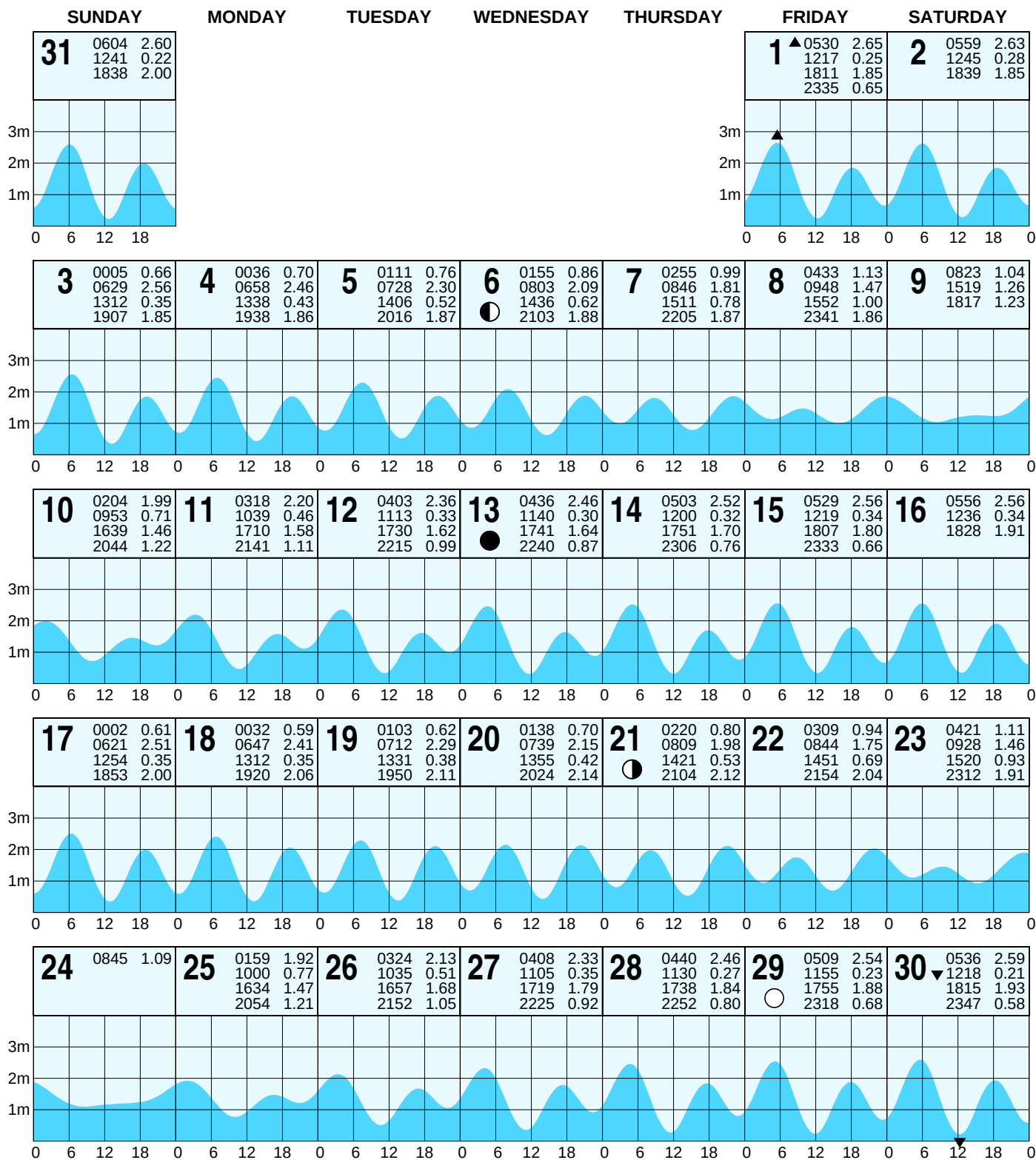
Notes: We've gone back to our standard color colorscheme

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month
▼ Lowest tide of the month

● New moon
◐ First quarter
○ Full moon
◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.8 Example 8. ..add shading for the night-time period between sunset and sunrise

CLASSIFICATION: UNRESTRICTED

Command: `tide_calendar2 -X -c 2 -f -s <year> <antt>`

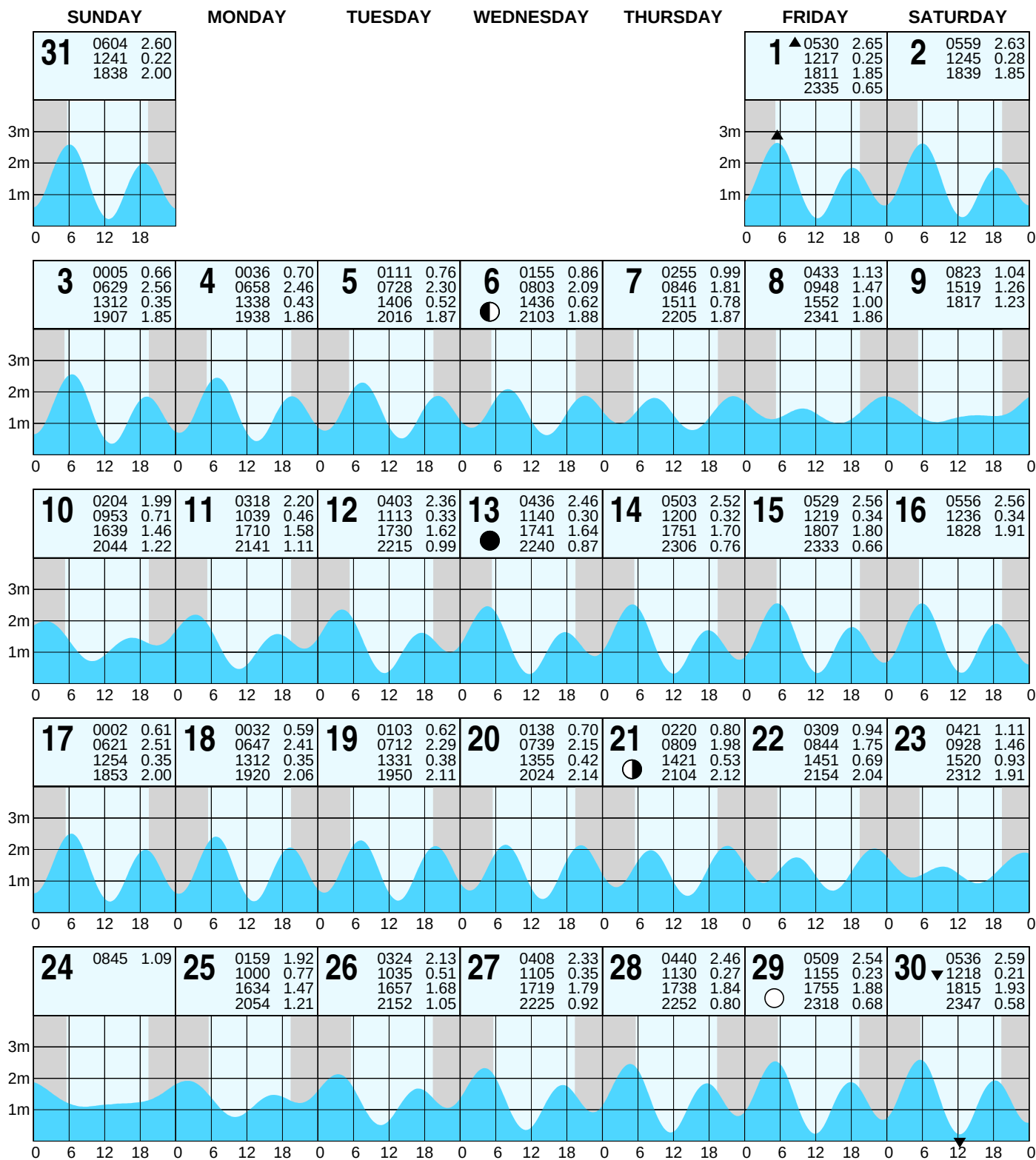
Notes: an upgrade to Gnuplot 4.4 would allow transparency, such that the night-time shading would be partly visible beneath the tide curve.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



- ▲ Highest tide of the month
- ▼ Lowest tide of the month
- Night time

- New moon
- ◐ First quarter
- Full moon
- ◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.9 Example 9. ..add a line tracing the tide curve

CLASSIFICATION: UNRESTRICTED

Command: `tide_calendar2 -X -c 2 -f -s -l <year> <antt>`

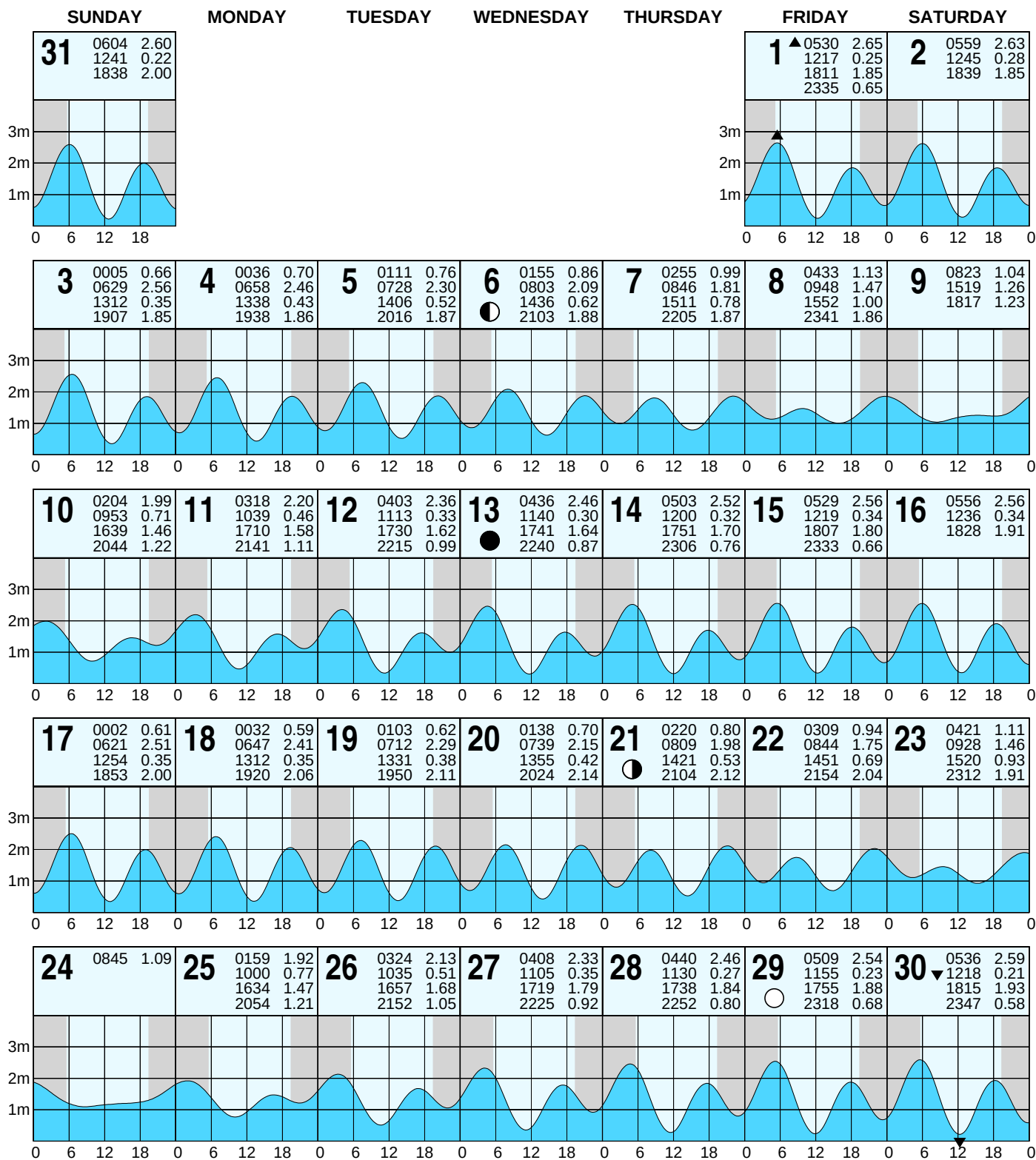
Notes: Generally looks better without, but a line can be used when a little more definition to the tide curve is needed to de-lineate between certain color-schemes

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month
 ▼ Lowest tide of the month
 Night time

● New moon
 ☾ First quarter
 ○ Full moon
 ☾ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.10 Example 10. ..highlight tides above/below certain thresholds

CLASSIFICATION: RESTRICTED

Command: `tide_calendar2 -X -c 2 -f -s -l -w 2.0 -d 0.5 <year> <antt>`

Notes: `tide_calendar2` has the capability to highlight tides that extend above or below chosen height thresholds (in this example 1.0m and 0.3m)

This feature could be adapted to a tidal datum such as above/below Mean Sea Level or above/below Mean High Water Springs/Mean Low Water Springs.

An example of this feature is http://tidesandcurrents.noaa.gov/tide_predictions.html

Although the capability exists, it is problematic to some degree in that different users may have different requirements.

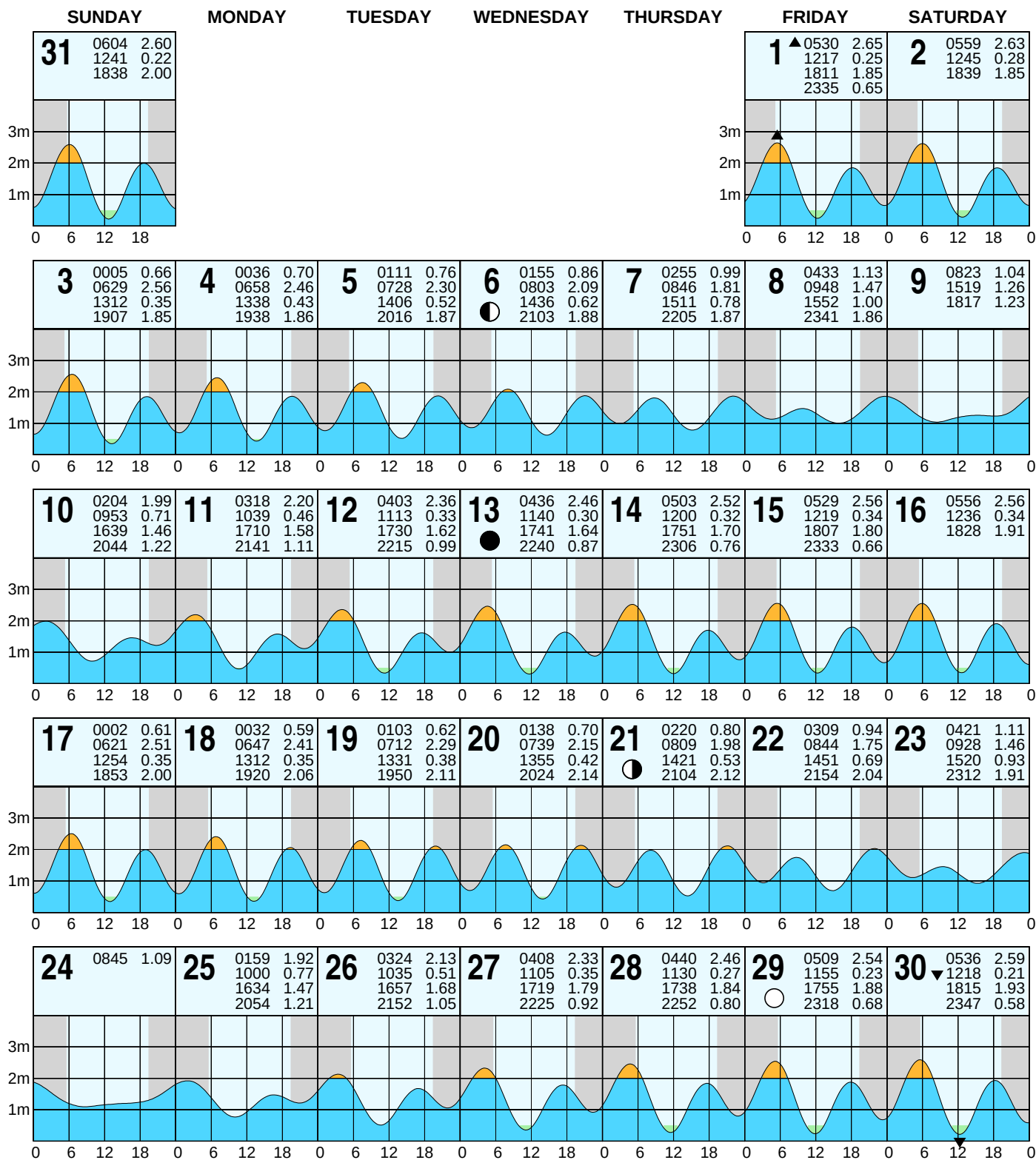
This capability is restricted from public release at this stage.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month

▼ Lowest tide of the month

Orange Tides above 2.0m

Green Tides below 0.5m

Grey Night time

● New moon

◐ First quarter

○ Full moon

◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.11 Example 11. ..add highlighting of tide times above/below certain thresholds

CLASSIFICATION: RESTRICTED

Command: *tide_calendar2 -X -c 2 -f -s -l -W 2.0 -D 0.5 <year> <antt>*

Notes: *tide_calendar2* has the capability to highlight tide times/heights that extend above or below chosen height thresholds (in this example 1.0m and 0.3m, but could be adapted to a tidal datum such as above/below Mean Sea Level or above/below Mean High Water Springs/Mean Low Water Springs).

An example of this feature is http://www.maritime.nsw.gov.au/docs/Tide_Tables.pdf

Although the capability exists, it is problematic to some degree in that different users may have different requirements.

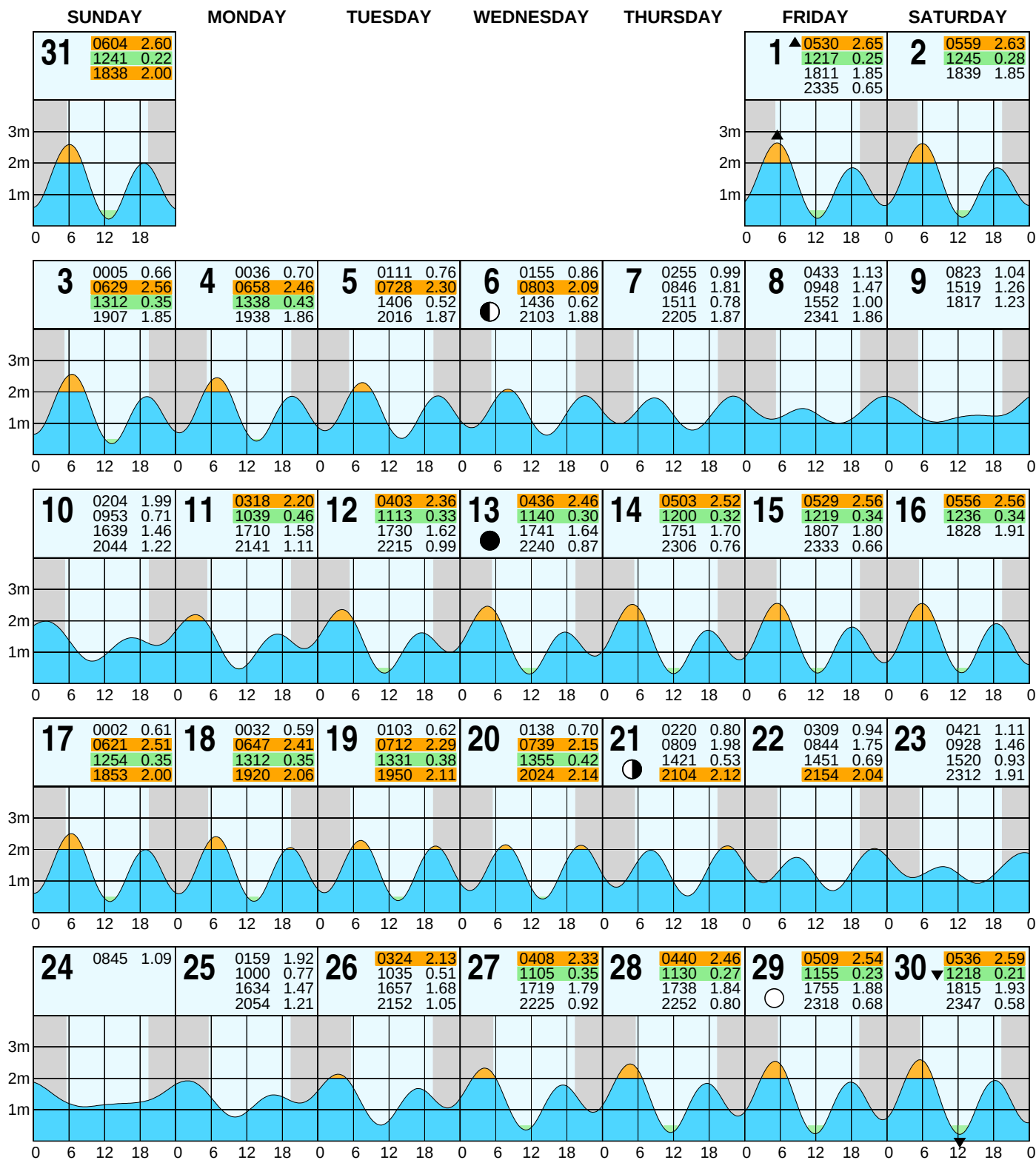
This capability is restricted from public release at this stage.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



- ▲ Highest tide of the month
- ▼ Lowest tide of the month
- Tides above 2.0m
- Tides below 0.5m
- Night time
- New moon
- ◐ First quarter
- Full moon
- ◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.12 Example 12. ..float the key/legend

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2 -X -c 2 -f -s -l -W 2.0 -D 0.5 -F <year> <antt>*

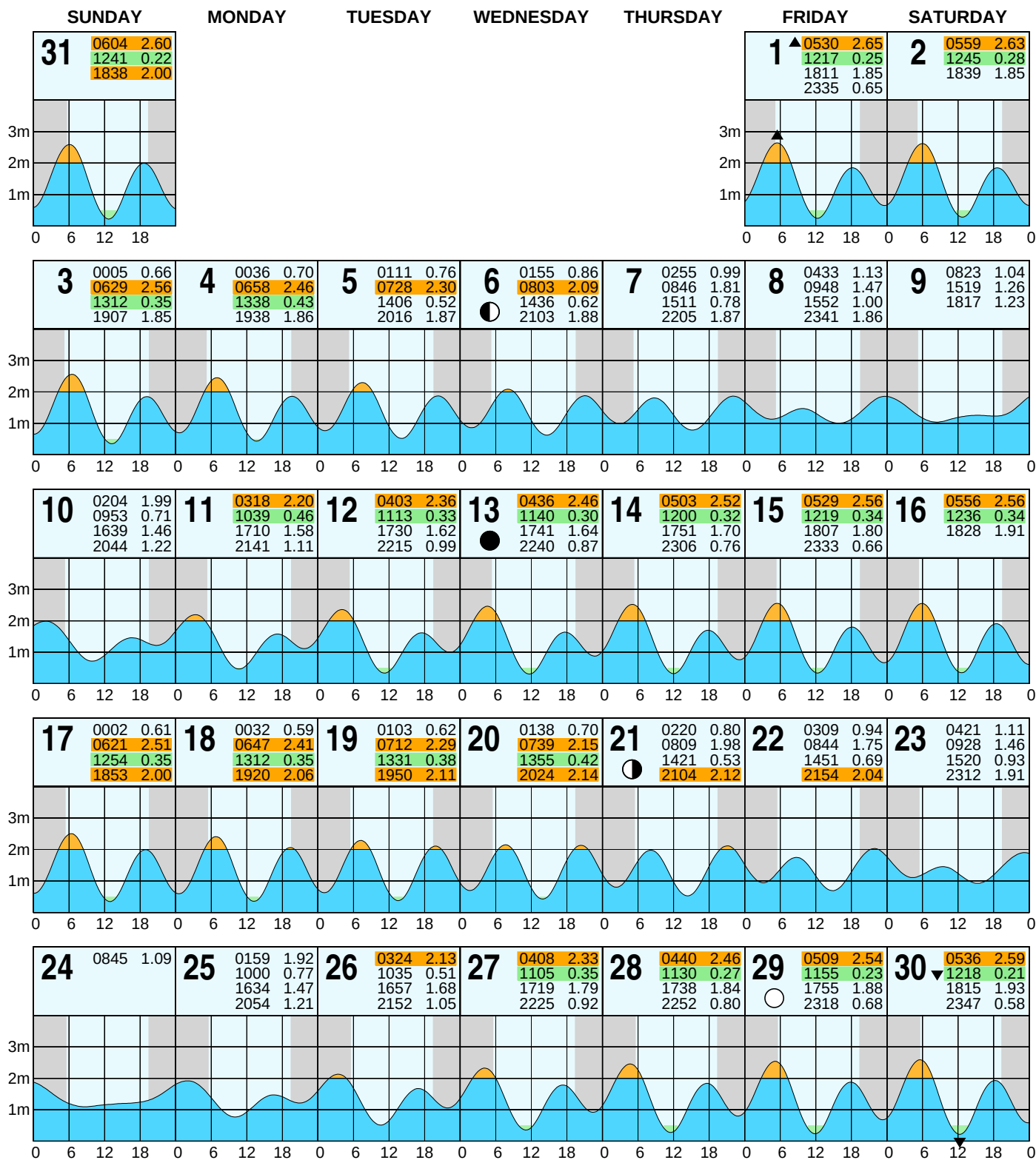
Notes: When the footer becomes too congested for the key, float it around from month-to-month in the vacant space in the main calendar area.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month

▼ Lowest tide of the month

Orange Tides above 2.0m

Green Tides below 0.5m

Grey Night time

● New moon

◐ First quarter

○ Full moon

◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.13 Example 13. Produce the Pacific Sea Level Monitoring project calendars

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2 -p pslm <year> <antt>*

Notes: The preset *pslm* tide calendars were designed in 2014 in conjunction with the PSLM Project Manager and COSPPac Communications and Capacity Development Staff in conjunction with graphic design assistance from Bureau publications.

pslm: Use this type for the hardcopy calendars. The color is black and white for reduced printing costs and page ordering that facilitates a double-sided top-bound calendar (*-o* option is hardwired in)

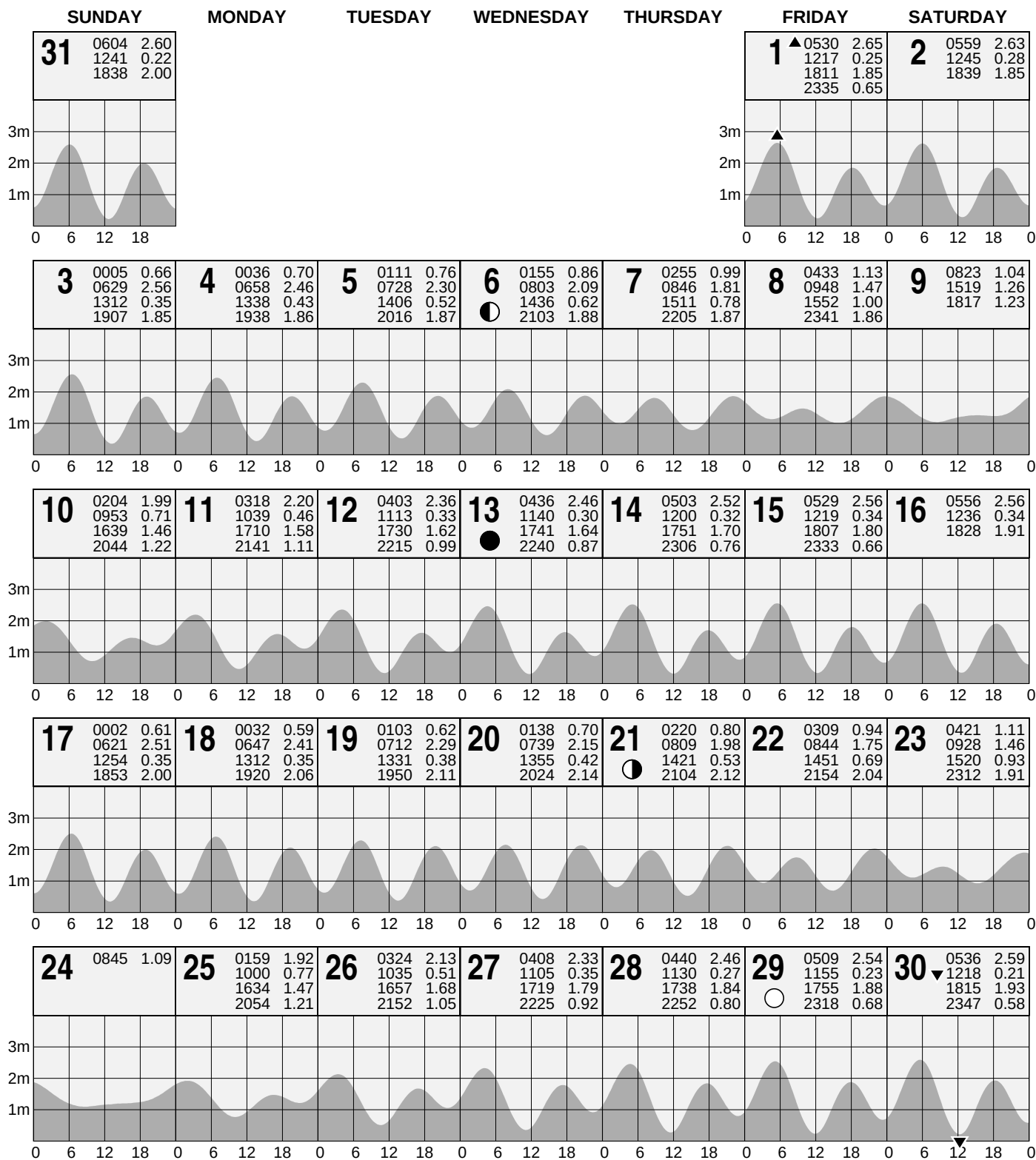
pslm_web: Use this type for the PSLM calendars on the website. They are essentially the same as *pslm*, but with linear page ordering rather than the double-sided top-bound option.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Add 1 hour to the predicted times during periods of Daylight Saving



▲ Highest tide of the month
 ▼ Lowest tide of the month

● New moon
 ◐ First quarter
 ○ Full moon
 ◑ Last quarter

© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and are believed to be correct. They are not necessarily related to a local hydrographic chart datum.

No warranty is given in respect to errors, omissions, or suitability for any purpose.

3.14 Example 14. Basic classic calendar

CLASSIFICATION: UNRESTRICTED

Command: *tide_calendar2 -t cal2 <year> <antt>*

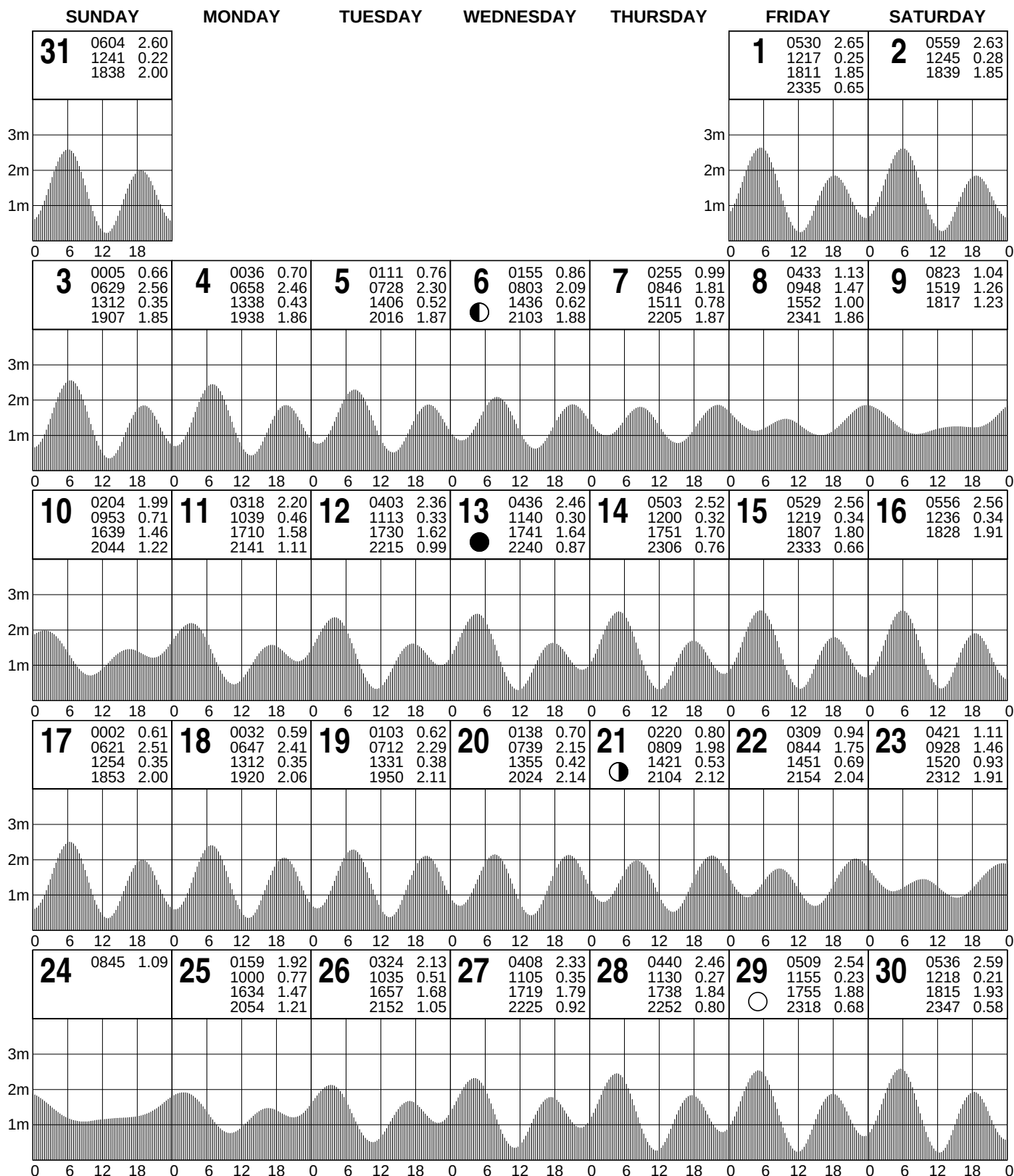
Notes: This was the original classic calendar before development of contemporary design (cal1) for PSLM project.

TIDAL PREDICTIONS FOR PORT ADELAIDE (OUTER HARBOR)

JANUARY 2021

Local Standard Time

Prediction Datum is Lowest Astronomical Tide



© Copyright: Commonwealth of Australia 2020, Bureau of Meteorology

Disclaimer: These tide predictions are supplied in good faith and believed to be correct.

No warranty is given in respect to errors, omissions, or suitability for any purpose.